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12/11/2019

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-387(388)/2019-001-01
UNIT 1 LICENSE NO. NPF-14
UNIT 2 LICENSE NO. NPF-22
PLA-7827

Docket No. 50-387
and 50-388

Attached is a supplement to Licensee Event Report (LER) 50-387(388)/2019-001-00. The LER reported a condition in which there was a loss of both the A and B Control Structure Chillers. This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v)(A), (B), (C), and (D) as a condition that could have prevented the fulfillment of a safety function. There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to be "K. Cimorelli", written over a horizontal line.

K. Cimorelli

Attachment: LER 50-387(388)/2019-001-01

Copy: NRC Region I
Ms. L. H. Micewski, NRC Sr. Resident Inspector
Ms. S. Goetz, NRC Project Manager
Mr. M. Shields, PA DEP/BRP

NRC FORM 366 (04-2018)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov , and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.	
 LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)					
1. Facility Name Susquehanna Steam Electric Station Unit 1			2. Docket Number 05000387		3. Page 1 OF 3
4. Title Loss of Both Control Structure Chillers due to Misaligned Breaker Stab					
5. Event Date			6. LER Number		7. Report Date
Month	Day	Year	Year	Sequential Number	Rev No.
05	22	2019	2019	001	01
			8. Other Facilities Involved		
			Facility Name Susquehanna Steam Electric Station Unit 2		Docket Number 05000388
			Facility Name		Docket Number 05000
9. Operating Mode 1		11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
		☐ 20.2201(b)		☐ 20.2203(a)(3)(i)	
		☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)	
		☐ 20.2203(a)(1)		☐ 20.2203(a)(4)	
		☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)	
		☐ 20.2203(a)(2)(ii)		☐ 50.73(a)(2)(ii)(A)	
		☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(ii)(B)	
		☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(iii)	
		☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(iv)(A)	
		☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(v)(A)	
		☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(v)(B)	
				☒ 50.73(a)(2)(v)(C)	
				☒ 50.73(a)(2)(v)(D)	
				☐ 50.73(a)(2)(vii)	
				☐ 73.71(a)(4)	
				☐ 73.71(a)(5)	
				☐ 73.77(a)(1)	
				☐ 73.77(a)(2)(ii)	
				☐ 73.77(a)(2)(iii)	
		☐ Other (Specify in Abstract below or in NRC Form 366A)			
12. Licensee Contact for this LER					
Licensee Contact Shane Jurek, Senior Engineer – Nuclear Regulatory Affairs				Telephone Number (Include Area Code) (570) 542-1695	
13. Complete One Line for each Component Failure Described in this Report					
Cause	System	Component	Manufacturer	Reportable To ICES	
B	ED	52	N140	Y	
14. Supplemental Report Expected					15. Expected Submission Date
☐ Yes (If yes, complete 15. Expected Submission Date)					☒ No
Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)					
On May 22, 2019, at 0240, the A Control Structure (CS) Chiller tripped due to a loss of power to Motor Control Center (MCC) 0B136. At the time of the trip, the B CS Chiller was out of service for planned maintenance on the Emergency Service Water (ESW) System. The CS Chillers are required to be operable by Technical Specifications (TS) Limiting Conditions for Operation (LCOs) 3.7.3, "Control Room Emergency Outside Air Supply (CREOAS) System" and 3.7.4, "Control Room Floor Cooling System." The CREOAS and Control Room Floor Cooling systems provide cooling and ventilation to various safety-related loads including the control room, computer room, relay rooms, and emergency switchgear rooms. The loss of both chillers required entry in LCOs 3.7.3, 3.7.4, and 3.0.3, thereby requiring the commencement of a dual unit shutdown. As a result, this event was reported in accordance with 10 CFR 50.72(b)(3)(v) as a condition which could have prevented the fulfillment of a safety function and 10 CFR 50.72(b)(2)(i) for the initiation of a plant shutdown required by TS. The power reduction was halted upon restoration of the A CS Chiller. The condition was also determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v) as a condition that could have prevented the fulfillment of a safety function. The direct cause was determined to be inadequate engagement of cubicle stabs. The root cause was determined to be vendor design differences in cubicle stab assemblies introduced during replacement of original plant equipment. Corrective actions include removing the misaligned breaker stab from service and de-energizing, cleaning, and inspecting MCC 0B136. There were no actual safety consequences associated with the condition.					

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Susquehanna Steam Electric Station Unit 1		05000-387	YEAR	SEQUENTIAL NUMBER	REV NO.
			2019	001	01

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

At the time of the event, Division II of the ESW System was out of service for planned maintenance. This rendered the B CS Chiller inoperable.

EVENT DESCRIPTION

On May 19, 2019, at approximately 2230, Division II of the ESW system was removed from service for planned maintenance. This rendered the B CS Chiller inoperable. Subsequently, on May 22, 2019, at 0240, the A CS Chiller [EIS Code: VI] tripped due to a loss of power to MCC 0B136 [EIS Code/Component Code: ED/MCC]. Susquehanna Steam Electric Station TS LCO 3.7.3, "Control Room Emergency Outside Air Supply (CREOAS) System," and LCO 3.7.4, "Control Room Floor Cooling System," require entry into LCO 3.0.3 upon loss of both CS Chillers. At 0256, operators declared the CREOAS and Control Room Floor Cooling Systems inoperable and entered LCO 3.0.3. Unit shutdown commenced at 0340 and 0350 for Units 2 and 1, respectively.

The MCC 0B136-044 A-phase cubicle stab [Component Code: 52] was not engaged correctly on the vertical bus resulting in arcing conditions that created the conditions for a fault on the MCC bus. The bus was tested with satisfactory results and was reenergized. Power was restored to MCC 0B136 at 0901 on May 22, LCO 3.0.3 was exited, and the power reductions were halted.

The CS Chillers are required to be operable to support operation of the CREOAS and Control Room Floor Cooling systems. The CREOAS and Control Room Floor Cooling systems provide cooling and ventilation to various safety-related loads including the control room, computer room, relay rooms, and emergency switchgear rooms. As a result, the condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(v) as a condition that could have prevented the fulfillment of a safety function of systems that are required to; (A) shut down the reactor and maintain it in a safe shutdown condition; (B) remove residual heat; (C) control the release of radioactive material; or (D) mitigate the consequences of an accident.

CAUSE OF EVENT

The direct cause of the event was determined to be that the 0B136-044 A-phase primary disconnect cubicle stab was not engaged on both sides of the bus bar. The root cause of the event was determined to be vendor design differences between the previous Cutler Hammer and replacement Nuclear Logistics Industries (NLI) assemblies affected the self-aligning capability which ensure cubicle stab engagement on the bus bar. On October 4, 2019, in accordance with 10 CFR 21, AZZ Nuclear notified the NRC of a

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CONTINUATION SHEET**

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			2019	001	01

deviation affecting F10 series MCC cubicles supplied by NLI with the current design of stab assembly P/N: 5600-SA3-1.

ANALYSIS/SAFETY SIGNIFICANCE

Due to the concurrent loss of the A and B CS Chillers, neither division of the Control Room Floor Cooling and CREOAS systems were available to perform their safety functions in the event of a design basis accident (DBA) (e.g., Loss of Coolant Accident Coincident with a Loss of Offsite Power). Any operator action to restore either CS Chiller would have required significant time and, therefore, would not have supported the safety function of the systems.

Based on engineering analysis, this event will be counted as a safety system functional failure under the NRC Reactor Oversight Process Performance Indicator.

CORRECTIVE ACTIONS

The load on Breaker 44 from MCC 0B136 was moved to a different breaker location within MCC 0B136. Additional corrective actions will include de-energizing, cleaning, and inspecting the entirety of MCC 0B136; and revising the procedures for NLI 480 VAC cubicle installation and maintenance to require verification of proper stab alignment.

COMPONENT FAILURE INFORMATION

Component Identification – 0B136044

Component Name – Control Structure Passenger Elevator Disconnect Switch 0DS108 Breaker

Manufacturer – Nuclear Logistics Industries

Model Number – F10 MCC Cubicle

Breaker Rating – 480 VAC

PREVIOUS OCCURRENCES

LER 50-388(387)/2015-015-00, "Loss of Safety Function due to Inoperability of Both Trains of the Control Room Emergency Outside Air Supply (CREOAS) System," dated May 5, 2016

LER 50-387(388)/2014-009-00, "Loss of Both Trains of Control Structure Chilled Water during Application of Clearance Order," dated August 11, 2014

LER 50-387(388)/2014-008-00, "Loss of Both Trains of Control Structure Chilled Water due to Personnel Error," dated August 11, 2014